Module 2 – Data Analysis

1. NumPy Basics

What is NumPy?

NumPy is a Python library used for numerical computing and working with arrays.

Why NumPy?

• Faster than standard Python lists
• Supports multi-dimensional arrays
• Provides mathematical functions

Example:

```
import numpy as np

arr = np.array([1, 2, 3])
print(arr)
```

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2. Pandas Introduction

What is Pandas?

Pandas is a powerful library used for data manipulation and analysis.

Key Features

• DataFrame structure
• Easy data handling
• Supports large datasets

Example:

```
import pandas as pd

data = {"name": ["Ali", "Sara"], "age": [20, 22]}
df = pd.DataFrame(data)
print(df)
```

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3. Data Cleaning

What is Data Cleaning?

Data cleaning is the process of fixing or removing incorrect, missing, or duplicate data.

Common Tasks

• Handling missing values
• Removing duplicates
• Correcting errors

Example:

```
df.dropna()
df.drop_duplicates()
```

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4. Exploratory Data Analysis (EDA)

What is EDA?

EDA is used to analyze datasets and summarize their main characteristics.

Techniques

• Summary statistics
• Data visualization
• Pattern detection

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5. Data Visualization

What is Data Visualization?

Visualization represents data using graphs and charts.

Popular Libraries

• Matplotlib
• Seaborn

Example:

```
import matplotlib.pyplot as plt

plt.plot([1,2,3], [4,5,6])
plt.show()
```

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6. Key Concepts to Remember

• NumPy handles numerical data
• Pandas manages datasets

• Cleaning improves data quality
• EDA explores data
• Visualization shows insights

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7. Summary

Data analysis is a crucial step in data science. Using tools like NumPy and Pandas, along with cleaning and visualization techniques, helps extract meaningful insights from raw data.